reconstruction of an hourly cooling-demand target from fragmented cumulative measurements. Third, forecasting literature supports comparative time-series modelling under strong autoregressive structure, weather sensitivity and horizon-dependent behaviour. Fourth, predictive-control and benchmark-oriented evaluation literature show that model outputs can be used not only for prediction, but also for supervisory decision logic and cautious historical comparison. The specific contribution of the present thesis lies in combining these strands within one real telecom-site case under realistic sensing constraints.

2. RESEARCH DESIGN, DATA AND METHODS

2.1 Research design and analytical workflow

This thesis adopts an applied quantitative research design based on retrospective analysis of historical monitoring telemetry from a selected telecom central office. The study is empirical and observational. It analyses previously recorded operational data rather than data generated through controlled experiments or live intervention. This design is appropriate because the objective of the thesis is to develop predictive models for cooling energy use and to evaluate intelligent control strategies under real operational constraints, not to validate new physical equipment or conduct actuator-level control experiments. In this sense, the research addresses cooling energy use as a real-world operational problem in a mission-critical technical environment.

The context of the study plays an essential role in the research design. Telecom central offices are continuously cooled environments in which thermal stability is required for the reliable operation of sensitive electronic equipment. At the same time, the available data are not structured as a ready-made analytical dataset. Instead, the research must begin from imperfect monitoring telemetry characterized by fragmented subsystem measurements, uneven completeness, and limited direct thermal-state information. For this reason, the study is designed as a single-site case study that prioritises methodological depth, traceability, and operational realism over broad but shallow cross-site comparison.

The design is structured around three linked analytical questions. The first concerns target engineering. How real operational and environmental data can be analysed and transformed into a defensible hourly cooling-demand target and analytical dataset. The second concerns forecasting - which modelling approach provides the most reliable short-term benchmark for expected cooling demand. The third concerns intelligent control. To what extent forecast-informed intelligent control strategies, evaluated through historical simulation, can identify conservative opportunities to improve cooling energy use. These questions are not independent. Each later stage of the study depends on the validated outcome of the previous one, and the research design therefore follows a sequential analytical logic rather than a set of isolated experiments.

This structure is consistent with the research gap identified in the literature review. The reviewed telecom literature is comparatively strong in monitoring, KPI development, and diagnosis, but weaker in the full chain from target reconstruction to short-horizon cooling forecasting, benchmark-based operational assessment, and supervisory control logic under sensing constraints.

The contribution of the thesis is therefore not the proposal of a new control algorithm or a new forecasting architecture in isolation. Rather, it is the development of a methodologically integrated applied framework that combines target engineering, comparative forecasting, benchmark-based evaluation, and intelligent control simulation within one real telecom central office case.

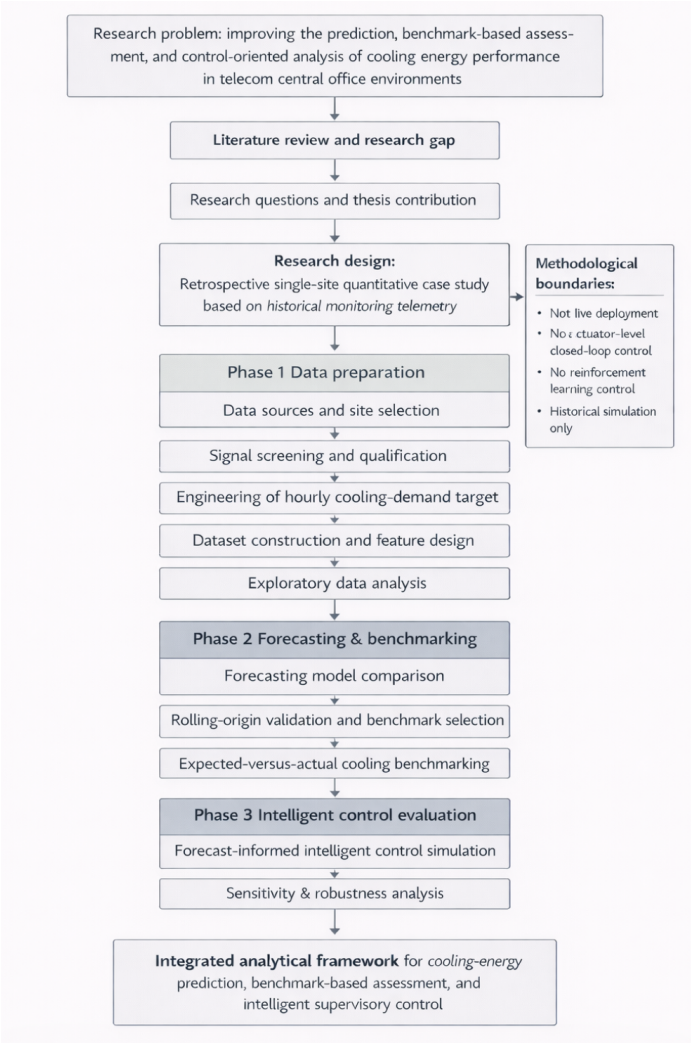

Figure 2.1 Research design and analytical workflow of the thesis

The overall analytical workflow of the thesis is summarised in Figure 2.1. The workflow begins with source review and selection of a suitable telecom site. It then proceeds to signal screening and qualification, engineering of an hourly cooling-demand target from fragmented cumulative subsystem telemetry, dataset construction and feature design, and exploratory data analysis. On that basis, short-term forecasting models are developed and compared under chronological and rolling-origin validation in order to select a defensible expected-demand benchmark. The selected benchmark is then used for expected-versus-actual cooling assessment and for the simulation of forecast-informed intelligent control strategies. The final stage consists of sensitivity and robustness analysis of the supervisory design.

Although the workflow is primarily data-driven, it is not purely automatic. Engineering judgement enters at several points, particularly in site selection, signal qualification, target-component review, and interpretation of supervisory operating modes and control states. This is especially important because the cooling target is not directly available as a ready-made hourly variable. It must first be reconstructed from fragmented cumulative subsystem measurements before forecasting, benchmarking, or intelligent control evaluation can be justified. For this reason, preprocessing and target engineering are treated in this thesis as substantive parts of the research design rather than as hidden technical preparation.

Within this design, predictive modelling serves two functions. The first is comparative and methodological: to determine which model family best captures short-horizon cooling demand in the selected context. The second is operational: to generate an expected-cooling benchmark against which actual cooling behaviour can be interpreted. Forecasting accuracy is therefore not treated as an end in itself. It matters because the selected model becomes the analytical reference for later benchmark-based assessment and for the residual logic used in intelligent control simulation.

The intelligent control component of the thesis is implemented at the supervisory level. It does not represent live deployment, actuator-level industrial control, full model predictive control implementation, or reinforcement-learning control. Instead, the study evaluates forecast-informed intelligent control strategies through historical simulation. These strategies use forecast deviations, contextual variables, explicit control states, operating modes, and bounded reduction rules to test whether conservative intervention opportunities can be identified under real telemetry constraints. This scope is methodologically consistent with the available data and with the position of the thesis in the literature review, where supervisory control is treated as a defensible intermediate layer between pure forecasting and full closed-loop advanced control.

Overall, the research design can be described as a stage-gated analytical framework in which each phase produces the validated input for the next. The study moves from raw monitored data to operational insight without assuming that either the target variable or the control logic already exists in ready-made form. Its contribution lies in bridging site-level telemetry, engineered cooling-demand reconstruction, comparative short-term forecasting, expected-versus-actual benchmarking, and forecast-informed intelligent control simulation within one coherent empirical study of cooling energy use in a telecom central office.

2.2 Study site and data sources

This section describes the database access context, the visible source tables, the basis for site selection, and the rationale for the selected study period.

2.2.1 Data access and extraction context

The empirical analysis was based on historical monitoring data obtained through read-only query access to a reporting database. Data extraction and inspection were performed using Python scripts. The first step was not full measurement loading, but an initial technical check of what was visible in the accessible reporting layer. A `SHOW TABLES` query confirmed that the visible tables were `datapoint`, `enmos_datapoint`, `measurement`, `measurement_all`, `measuring_point`, `mobnot`, `mobnot_reported_state`, `monitoring_site`, `monitoring_site_model`, and `swenex_file_data`.

After confirming the visible tables, a lightweight metadata inventory was created before any heavy time-series extraction was attempted. Column metadata from the visible tables were exported to `columns_metadata.csv`, and a separate helper file, `tables_relations.csv`, was used to keep track of practical links between selected sources, for example between `monitoring_site_model` and `monitoring_site`, `enmos_datapoint` and `monitoring_site`, and `swenex_file_data` and `monitoring_site`.

This access structure determined the order of the analytical workflow. Because the measurement layer was considerably heavier than the metadata layer, the study began with source discovery and metadata inspection rather than immediate bulk extraction of measurement records. In practical terms, `monitoring_site` and `monitoring_site_model` provided site-level context, `enmos_datapoint`, `datapoint`, and `measuring_point` described the available signals, `measurement` and `measurement_all` contained time-stamped values, and `swenex_file_data` provided supporting utility-related information used during initial screening.

This is important for the interpretation of the study data. The final analytical dataset was not available as one ready-made table. It had to be assembled progressively from several visible reporting sources, starting from table discovery, moving to metadata review, and only then proceeding to site selection, signal qualification, and later target construction.

2.2.2 Accessible source tables and their analytical roles

The accessible reporting layer was not a single modelling table, but a small relational source set consisting of ten visible tables: `monitoring_site`, `monitoring_site_model`, `swenex_file_data`, `enmos_datapoint`, `datapoint`, `measuring_point`, `measurement`, `measurement_all`, `mobnot`, and `mobnot_reported_state`. The initial discovery step confirmed these visible objects and a metadata export was created before large-scale extraction began. In parallel, a small practical relations file was used to keep track of the joins that were directly usable in the reporting layer, including `monitoring_site_model.monitoring_site_id -> monitoring_site.id`, `enmos_datapoint.site_code -> monitoring_site.site_code`, `swenex_file_data.code -> monitoring_site.site_code`, and `mobnot_reported_state.mobnot_id -> mobnot.mobnot_id`.

From an analytical point of view, the source layer can be divided into three main groups. The first group is the **site and support layer**, represented mainly by `monitoring_site`, `monitoring_site_model`, and `swenex_file_data`. `monitoring_site` provided the central site identifier and path metadata. `monitoring_site_model` provided the link to the site through `monitoring_site_id` and, in the visible schema, also contained model- and context-level fields such as `schema_id`, `lat`, `lon`, `version`, and `doc`. The `doc` field contained a structured JSON site description, including technical groups such as cooling circuits, meteo, transformers, rectifiers, utility meters, and ventilation subsystems. In the thesis, this document is treated as a technical reference for semantic interpretation of the telemetry, especially for target engineering and later control-context interpretation, not as a primary measurement source. Geographic coordinates from this source were used for external weather retrieval, but are not reported in the thesis for confidentiality reasons. `swenex_file_data` provided utility-related support information such as coverage period, total consumption, and cost fields used in the initial site-screening stage.

The second group is the **signal metadata layer**, represented by `enmos_datapoint`, `datapoint`, and `measuring_point`. `enmos_datapoint` provided site-level metadata fields such as `site_code`, `name`, `description`, `source_class`, `status`, `unit`, and `object_id`, and it was used for keyword-based screening of candidate sites and later for comparison against the datapoint-mapped signal universe. `datapoint` provided the main signal registry used in the downstream experiment, with fields such as `id`, `measuring_point_id`, `dp_id`, `type`, `name`, `description`, `vendor`, `device_type`, `quantity`, and `uom`. `measuring_point` added structural context for the datapoints through fields such as `mp_id`, `type`, `name`, `description`, `vendor`, `device_type`, and `path`. In

practical terms, datapoint.id was the key technical field connecting the metadata layer to the measurement layer, while datapoint.dp_id was also important because it was used to compare datapoint records with ENMOS names during cross-table mapping.

The third group is the **measurement layer**, represented by measurement and measurement_all. measurement is the main normalized time-series table used in the thesis, built around datapoint_id, time, and value. It became the core source for signal profiling, hourly aggregation, target construction, and final dataset assembly. measurement_all is a broader denormalized measurement view that includes time, value, time_local, datapoint_id, dp_id, site_code, and duplicated signal and measuring-point metadata such as name, description, vendor, device_type, mp_name, mp_type, mp_description, quantity, uom, and mp_id. In the experiment, measurement_all was used mainly for source-consistency checks against measurement and for exploratory lookup of signals that were visible in ENMOS metadata but not directly mapped in the datapoint table. That distinction became methodologically important later, because the final experiment remained anchored on the datapoint-mapped signal set rather than on ENMOS-only names. The remaining visible tables, mobnot and mobnot_reported_state, were part of the accessible reporting environment but were not used in the final analytical workflow. Table 2.1 summarises the source tables and their roles in the study.

Source table	**Main technical fields visible/used**	**Analytical role in the thesis**
monitoring_site	id, site_code, path	Central site registry used for site identification and linkage of site-level metadata
monitoring_site_model	id, monitoring_site_id, schema_id, lat, lon, doc, version	Site-model table used for site linkage, weather-coordinate source, and technical semantic interpretation through the structured doc field
swenex_file_data	code, location_type, customer, city, canton, street, house_number, from_date, to_date, total_kwh_consumption, total_cost, energy/network/fee fields	Utility-support table used during candidate-site screening and coverage assessment

enmos_datapoint	id, site_code, name, description, source_class, status, unit, object_id, kafka_export, sdbm_since	Site-level signal metadata used for keyword-based screening and ENMOS-to-datapoint comparison
datapoint	id, measuring_point_id, dp_id, type, name, description, vendor, device_type, quantity, uom	Main signal metadata table used to define the candidate signal pool and connect signals to measurements
measuring_point	id, mp_id, type, name, description, vendor, device_type, path, monitored_id	Additional structural context for datapoints and source organisation
measurement	datapoint_id, time, value	Main normalized time-series source used for signal profiling, hourly aggregation, target engineering, and dataset construction
measurement_all	time, value, time_local, datapoint_id, dp_id, site_code, name, description, vendor, device_type, mp_name, mp_type, mp_description, quantity, uom, mp_id, path, type	Denormalized measurement view used for source-consistency checks and exploratory lookup of ENMOS-only or loosely mapped signals
mobnot	id, mobnot_id, inventory_id, path, name, description, vendor, device_type, status/range fields	Visible in the reporting layer, but not used in the final analytical workflow

mobnot_reported_state	id, mobnot_id, doc, created_at	Visible in the reporting layer, but not used in the final analytical workflow

In practice, the final thesis workflow depended mainly on the interaction between the site/support layer, the signal metadata layer, and the measurement layer; the two mobnot tables remained outside the final experiment design.

2.3 Signal screening and variable-selection logic

2.4 Engineering of the hourly cooling-demand target

2.4.1 Selection of cumulative subsystem signals
2.4.2 Hourly alignment and differencing
2.4.3 Treatment of negative or implausible steps
2.4.4 Aggregation into final target y

2.5 Dataset construction and feature design

2.5.1 Main dataset
2.5.2 Compare dataset
2.5.3 Weather and calendar variables
2.5.4 Operational variables
2.5.5 Internal temperature proxy

2.6 Exploratory data analysis
2.7 Forecasting methodology and validation design

2.7.1 Benchmark models
2.7.2 Machine-learning models
2.7.3 Lag structure
2.7.4 Rolling-origin validation
2.7.5 Evaluation metrics

2.8 Benchmarking and supervisory-control methodology

2.8.1 Expected-versus-actual cooling logic
2.8.2 Residual interpretation and reduction funnel
2.8.3 Supervisory strategy family B0–S3